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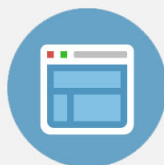
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Bond breaking without barriers: Photofragmentation of ketene at the singlet threshold

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Ketene (CH_2CO) in a supersonic free jet was photodissociated by a tunable pulsed laser in the frequency range just above the threshold for production of singlet methylene, CH_2 ($\tilde{a}^1A_1$). CH_2 was detected by laser-induced fluorescence (LIF). The appearance threshold and yield curve of individual $^1\text{CH}_2$ rotational states were obtained by scanning the photolysis laser frequency with a fixed LIF probe laser frequency. The dissociation occurs on the ground electronic state potential energy surface. The threshold for $\text{CH}_2\text{CO} \rightarrow ^1\text{CH}_2 + \text{CO}$ is found to be $30\,116.2 \pm 0.4\text{ cm}^{-1}$. By varying the delay between the photolysis and probe pulses, a lower bound of $7 \times 10^7\text{ s}^{-1}$ was set for the dissociation rate on the triplet surface at the singlet energy threshold. The yield curves, or photofragment excitation (PHOFEX) spectra, exhibit sharp steps spaced by the CO rotational term values. The experimental data provide a rigorous test of theoretical models of photofragment dynamics. The data clearly show that nuclear spin is conserved through the photodissociation. PHOFEX curves calculated from phase space theory (PST) are in excellent agreement with the experiment and show that there is no barrier along the reaction coordinate. The singlet/triplet branching ratio as a function of photolysis laser frequency is inferred from PST fits to the PHOFEX data. Comparisons with the statistical adiabatic channel model (SACM) are also presented. The data show that the only dynamical constraints on product state rotational energy distributions are conservation of energy, angular momentum, and nuclear spin.

I. INTRODUCTION

The combined application of laser and molecular beam methods makes it possible to study unimolecular reactions with the initial state of the reactant fully defined and the final quantum states of the products fully resolved. Such experiments can provide significant qualitative understanding of the dynamics of unimolecular processes as well as discriminating quantitative tests of alternative dynamical theories. The well-established RRKM theory provides a qualitatively satisfying and quantitatively tested model for unimolecular reactions with well-defined potential energy barriers. A statistical calculation of the reaction rate of an excited molecule relies on the assumption that energy is transferred rapidly among the degrees of freedom of the molecule compared to its rate of reaction. The reaction occurs when the molecule randomly passes through the transition state configuration. For potential surfaces with barriers this transition state is well defined. However, when the potential increases smoothly from reactant to products without a barrier, a transition state is not clearly defined. Such is the case for the breaking of most chemical bonds. If the transition state is placed at infinite separation of the fragments, the statistical rate may be straightforwardly calculated (the phase space theory) but the assumption of rapid energy flow between the separating fragments becomes suspect. A successful dynamical model must not only define the appropriate transition state but also the extent to which energy is transferred among vibrational and rotational degrees of freedom as the bond stretches and breaks. Several models have been proposed including the Prior,¹ phase space theory (PST),^{2,3} statistical

adiabatic channel model (SACM),⁴⁻⁶ SACM-PST,⁷ and separate statistical ensembles (SSE)⁸ models, as well as new modifications of the classic RRKM theory.^{9,10}

The PST model is based on a simple physical idea: that all accessible product phase space is equally likely to be populated. Conservation of total energy and total angular momentum are the only constraints. PST has been applied with success to a variety of systems^{2,3,8,11-13}; however, it fails to correctly predict the partitioning of energy between rotation and vibration in some systems.^{8,13,14} PST is found to provide excellent fits to the data in the present work. Prior theory ignores angular momentum conservation and centrifugal barriers; total energy conservation is the only constraint. It has the advantage that its predictions are completely independent of the properties of the reactant and of the potential energy surface. It has been used with success in a variety of cases,^{1,15-17} but fails when the initial state of the reactant is well specified and the initial rotational energy is significantly less than the product translational energy. Prior also fails to reproduce effects due to the quantization of angular momentum. The SSE method is a modification of PST which constrains energy flow between vibration and overall rotation of the dissociating molecule. It is identical to PST for total energies below the threshold for formation of vibrationally excited products. It has been applied with excellent success to the rovibrational product state distribution of NCNO.^{8,13,14}

There has been frequent speculation that statistical theories such as PST may correctly predict product state distributions, but fail to correctly predict dissociation rates.^{6,18} This can occur if there is energy randomization between the

outgoing fragments in the exit channel, with the rate being determined by passage through a critical configuration for which the number of states is less than at infinite separation. Recent work by Khundkar *et al.*¹⁹ on NCNO using pico-second lasers shows that although PST correctly predicts the product state distribution, it overestimates the measured dissociation rates except very close to threshold.^{19,20} However, the NCNO dissociation rate may be limited by internal conversion rather than by some non-PST behavior in dissociation on the ground potential energy surface. In the present work on CH₂CO, the absolute dissociation rate is not measured directly; however, the rates calculated using PST are reported.

The SACM theory, and the related SACM-PST approximation, separates the vibrations of the products from the other degrees of freedom, and calls for additional barriers for the various dissociation channels. These effective barriers arise because each vibrational state of the dissociating molecule is assumed to *adiabatically* correlate with a specified rovibrational state of the fragments. Thus, part of the energy is constrained not to flow into the reaction coordinate. The barriers in SACM reduce the dissociation rate from that calculated using PST. SACM has been successfully applied to a wide variety of systems,^{4-6,20-25} including detailed results from the photodissociation of H₂O₂.^{7,11} However, PST apparently works better than SACM for predicting the product state distribution from the photodissociation of NCNO near threshold, a finding which has sparked controversy.^{18,26}

Unfortunately, there is a lack of decisive evidence favoring one theory over others. The important physical differences between the various models of unimolecular dissociation are often not apparent when they are tested against data measured at high temperatures and low resolution. However, as illustrated in the present work, striking differences among theories of unimolecular dissociation can become apparent when a nearly monoenergetic ensemble of reactant molecules is prepared in a supersonic free jet near the reaction threshold.

The dissociation of ketene to form methylene has been studied for decades because of the importance of methylene as a model system for understanding singlet-triplet interactions and reactivity, and because of the recently resolved controversy over the size of the singlet-triplet energy gap in methylene.²⁷ The present work adds to knowledge of the thermochemistry, dissociation rate, and singlet-triplet branching ratio of this important photochemical system.

The low-lying electronic energy states of ketene are illustrated in Fig. 1. The figure reflects the results of recent *ab initio* calculations.^{28,29} After excitation to the $\tilde{A}^1A''$ surface, the molecule undergoes internal conversion or intersystem crossing to the lower-lying electronic surfaces, followed by dissociation. The absorption spectrum of ketene is diffuse and no fluorescence has been observed, presumably because of rapid relaxation from the $\tilde{A}$ state. The molecule has been studied by fixed frequency photolysis at 351 and 308 nm.^{16,30,31} At 308 nm the singlet channel dominates and the product state distribution is statistical as expected because there is no barrier.^{16,31} At 351 nm only the triplet channel is energetically accessible, and the photofragment state distribution

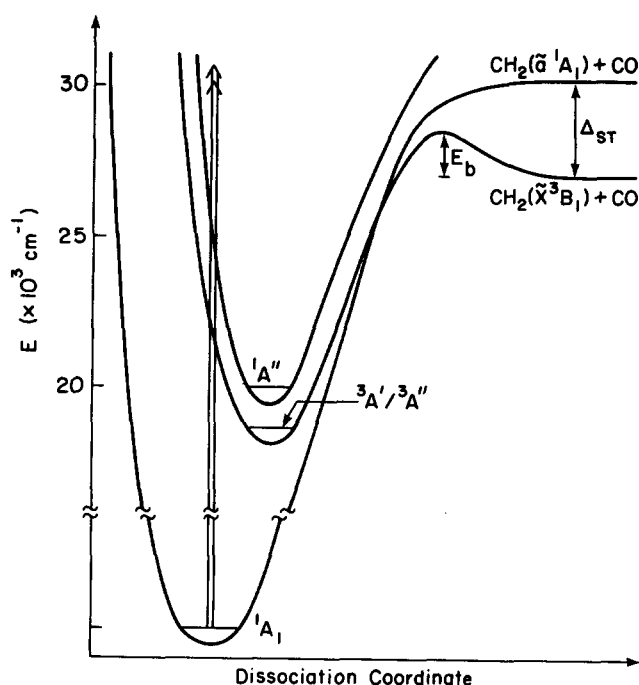


FIG. 1. The potential energy surfaces of the low-lying electronic states of ketene, adjusted for zero-point energy in the modes other than the reaction coordinate.

is nonstatistical indicating a small barrier (≤ 4.4 kcal/mol).^{30,31}

The PHOFEX technique, where the yield of one product state is measured as the photolysis wavelength is scanned, has been applied to ketene in the singlet threshold region in an earlier study, using photolysis wavelengths as short as 310 nm.¹⁵ The PHOFEX spectra obtained, which exhibited interesting fine structure, were approximately fit using the Prior theory.

In the present PHOFEX study on ketene, using a colder supersonic jet, the importance of angular momentum conservation becomes evident when trying to fit the well-resolved fine structure. This structure is sharpest in the threshold region, where fragment state distributions are expected to be most sensitive to the details of the photofragmentation dynamics.^{20,26,32} The present study was confined to photolysis wavelengths longer than 325 nm where the colder jet conditions provide a dramatic improvement over the previously published data. The more detailed PHOFEX data presented here provide a qualitatively better understanding of the photofragmentation dynamics.

II. EXPERIMENTAL

Ketene, cooled in a supersonic jet, was photolyzed by a tunable pulsed UV laser. The CH₂ ($\tilde{a}^1A_1$) produced was excited by a second pulsed dye laser tuned on a single ($\tilde{b}^1B_1 - \tilde{a}^1A_1$) rovibronic transition and the fluorescence was detected with a photomultiplier tube. The time delay between the photolysis and LIF probe pulses was varied to measure the appearance rate of $^1\text{CH}_2$.

Ketene was prepared by passing acetic anhydride through a red-hot quartz tube, was trapped at 77 K and distilled from 196 to 77 K twice. Prior to use the ketene was transferred to a U tube maintained at 179 K, hexanes (A.C.S. Reagent Grade) slush, where its vapor pressure is 50 Torr.³³ One atmosphere of He carrier gas was flowed over the ketene and into the pulsed nozzle (Newport; 1 or 0.5 mm orifice). The lower seeding ratio of ketene in He in the present work significantly reduced its thermal energy compared with previous work.¹⁵

The output of a Lambda-Physik FL2002 dye laser (0.2 cm^{-1} bandwidth) pumped by a Quanta Ray DCR-1A Nd:YAG laser was frequency doubled with a KDP crystal (inside an InRad Autotracking System) to produce the photolysis pulses. Mixtures of DCM and LDS698 dyes in methanol, and solutions of DCM dye in DMSO were used. The system produced tunable UV pulses from 325–335 nm (3–4 mJ/pulse). The second harmonic was separated from the fundamental using filters and dichroic mirrors. The beam cross section was $\approx 5 \text{ mm}^2$. The UV pulse energy fluctuated by about $\pm 30\%$ shot to shot.

For absolute photolysis frequency calibration using the optogalvanic effect, a portion of the mixed visible and UV beam was sent into an Fe:Ar hollow cathode lamp (Hamamatsu, $I < 10 \text{ mA}$). A portion of the visible beam was also monitored with an air-gap étalon of 2.5 cm^{-1} FSR. Both the étalon signal and the optogalvanic signal were recorded by a microcomputer simultaneously with the experimental data. The optogalvanic spectra were assigned using the MIT Wavelength Tables³⁴ and a more recent listing of the Fe spectral lines.³⁵ The position of every data point was interpolated between the étalon maxima, and converted to absolute frequency using the assigned optogalvanic lines. The overall error in the calibration procedure was found to be always less than 0.5 cm^{-1} in the ultraviolet. The frequency calibration procedure was important because of significant nonlinearities in the laser grating drive mechanism.

Two different probe laser systems were used, with virtually identical results: a Nd:YAG-pumped dye laser (grating-modified Quanta Ray PDL-1, 0.4 cm^{-1} bandwidth, pumped by a second DCR-1A), and a XeCl excimer-pumped dye laser (a second FL2002, pumped by a Lambda-Physik EMG103MSC). Kiton Red, R6G, and C540A dyes were used and the output energy was $< 30 \text{ mJ/pulse}$ with the YAG system, $< 10 \text{ mJ/pulse}$ with the excimer system. The probe laser cross sections were about 10 mm^2 in the interaction region. All of the dye lasers were run multimode, with a cavity mode spacing of about 0.02 cm^{-1} . The unfocused photolysis and probe laser beams were combined collinearly by a dichroic UV mirror and crossed the jet perpendicularly 4.0 cm downstream of the nozzle. The polarization of the electric field of the probe beam is parallel to the jet while that of the photolysis beam is perpendicular.

Fluorescence of CH_2 was detected perpendicularly to both the laser beams and the free jet by a red sensitive PMT (EMI 9658R). Long-pass cutoff filters (Schott KV520 and RG610/RG630) were used to filter out scattered light. Monts *et al.*³⁶ have reported a diffuse luminescence with a lifetime longer than $50 \mu\text{s}$ from multiphoton excitation of

ketene. Weak long lifetime luminescence ($< 5\%$ of signal intensity) while the probe beam is blocked was also observed in the present work. No variation in the long lifetime luminescence with excitation frequency was observed, except for a few sharp resonances. The fluorescence lifetime of singlet methylene is about $4 \mu\text{s}$.³⁷

The amplified signal from the PMT was sent into a gated integrator (Stanford Research Systems Model 250). The integrated output was stored by a microcomputer (DEC PDP 11/73 or Fountain PC XT) which also triggered and scanned the lasers and averaged signal for several shots at each data point. The photolysis laser power was measured using a photodiode and a second gated integrator. The LIF intensity of $^1\text{CH}_2$ was normalized to this signal. When the transitions of $^1\text{CH}_2$ were not saturated, the fluorescence intensity was normalized to the probe laser power as well.

Another gated integrator was used as a delay generator to change the time delay between photolysis and probe pulses, using the Nd:YAG pumped probe laser. The “Lamp-Sync-Out” pulses from one Nd:YAG laser triggered the gated integrator. The pulses from the “gate out” of the gated integrator were amplified and used to trigger the Q switch on the other Nd:YAG laser. An IBM PC microcomputer and SR265 software were used to externally vary the time delay between trigger and gate of the boxcar (via a computer interface, Stanford Research Systems Model 245) and to collect the data. The overall jitter in the delay was $\pm 4 \text{ ns}$. The duration of the output was about 5 ns from the Nd:YAG pumped dye lasers, and about 10 ns from the excimer-pumped dye laser. The excimer system could not be used for the time-resolved measurements because of the $\pm 40 \text{ ns}$ jitter between the trigger and the excimer laser output.

To ensure that the methylene transitions probed were not blended, the $^1\text{CH}_2$ LIF spectra in the region of interest were measured using a high resolution laser system described previously.¹⁵ These spectra were Doppler limited ($\approx 0.1 \text{ cm}^{-1}$). The LIF spectra were remeasured at a number of photolysis wavelengths between 330 and 325 nm to determine the approximate thresholds of weak transitions discovered to overlap the desired transition at the resolution used when recording the PHOFEX spectra.

In order to estimate the rotational temperature of ketene in the supersonic free jet, the LIF spectrum of D_2CO ($\tilde{A}-\tilde{X}$, $2_0^2 4_0^1$) was also measured using the high resolution laser system. D_2CO at 20 Torr was mixed with 1 atm He at dry ice temperature.³⁸ The 1 mm diameter nozzle orifice was used. The laser power was attenuated with neutral density filters to ensure that the line intensities were not distorted by saturation effects. In the jet, only $J'' < 2$ states were visible in the spectrum. The intensities were analyzed using asymmetric top line strengths³⁹ and the experimental fluorescence quantum yields⁴⁰ to give $T_{\text{rot}} = 2.6 \pm 0.8 \text{ K}$.

III. RESULTS

The PHOFEX spectra of several CH_2 ($\tilde{a} \ ^1A_1$) rotational states of the lowest vibrational state were measured, and are shown in Figs. 2–6 and Ref. 43. The ultraviolet absorption spectrum of ketene is almost structureless^{15,41,42} in the range

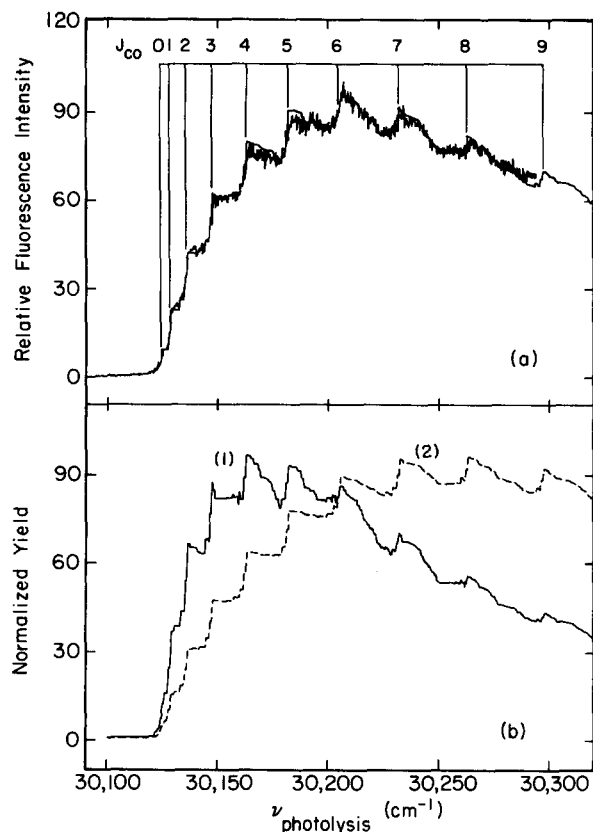


FIG. 2. (a) PHOFEX spectrum of the 1_{01} state measured with the probe laser fixed on the $\text{CH}_2(0,15,0)2_{11} \leftarrow (0,0,0)1_{01} (\bar{b} \leftarrow \bar{a})$ transition at $17\,757.76\text{ cm}^{-1}$. Nozzle aperture = 0.5 mm. The CO term values indicated match the experimental spacings. The PST fit to the data is also shown ($W' = 40$, $C_6 = \infty$, $b_{\text{max}} = \infty$, $T = 3\text{ K}$). (b) The effect of the triplet rate parameter W' , Eq. (14), on the PST fits for the 1_{01} state: (1) $W' = 10$ (solid), (2) $W' = 120$ (dashed).

studied here. The 0.2 cm^{-1} probe laser bandwidth is broader than the 0.1 cm^{-1} Doppler profile of the methylene and thus detection sensitivity is independent of fragment velocity. Hence, the structure of the PHOFEX spectrum is almost entirely due to changes in the quantum yield of the probed rotational state of CH_2 from ketene photodissociation as a function of photolysis energy.

All of the PHOFEX spectra show a rise from threshold as the probed channel opens, and then a decline at high photolysis energies as many additional singlet methylene states become energetically accessible and can compete with the specific rotational state being probed. In Fig. 2, the maximum occurs about 80 cm^{-1} above the experimental threshold of $30\,125.0\text{ cm}^{-1}$. The PHOFEX spectra show fine structure superimposed on these gross features. Sharp steps in the intensity are spaced by the term values of the other fragment (CO), $\Delta E = 2B_{\text{CO}}J$, $J = 1, 2, \dots$, Fig. 2. The fine structure is much sharper in the present work than in the report of Bitto *et al.*¹⁵ because colder ketene is employed (lower dilution ratio in He). These steps are expected to occur because many new product channels are switched on as each higher rotational state of CO becomes energetically accessible.¹⁵ The data in Figs. 2, 3, and 5 show that PHOFEX spectra of even the lowest rotational states 0_{00} and 1_{01} rise stepwise to a maximum. Since these steps occur below the threshold for the next higher singlet channel, there must be another competing decay channel for the excited ketene. Thus, the quantum yield of singlet methylene is significantly less than unity in the threshold region. The PST model presented below provides an understanding of all of the qualitative features of the data as well as a quantitative fit to the spectra.

A lower limit was set for the dissociation rate of ketene by varying the time delay between the photolysis and probe

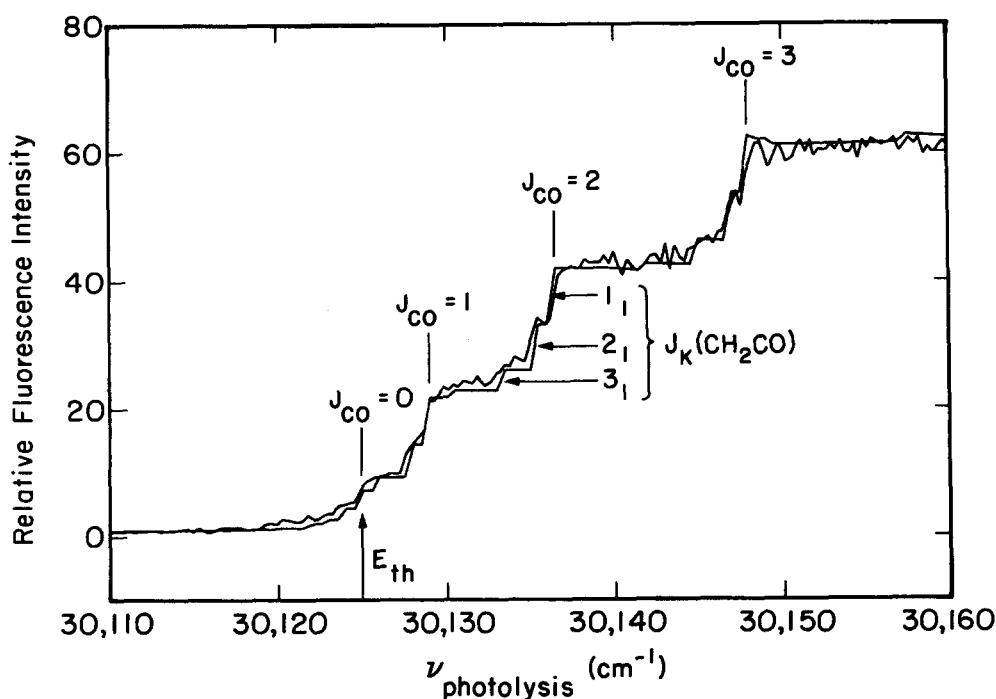


FIG. 3. The threshold region of the PHOFEX spectrum shown in Fig. 2, indicating the accuracy with which the threshold can be determined by the PST fit. The thresholds of different CO rotational state channels and the small steps due to the initial rotational distribution of ketene in the jet are indicated on the PST fit curve.

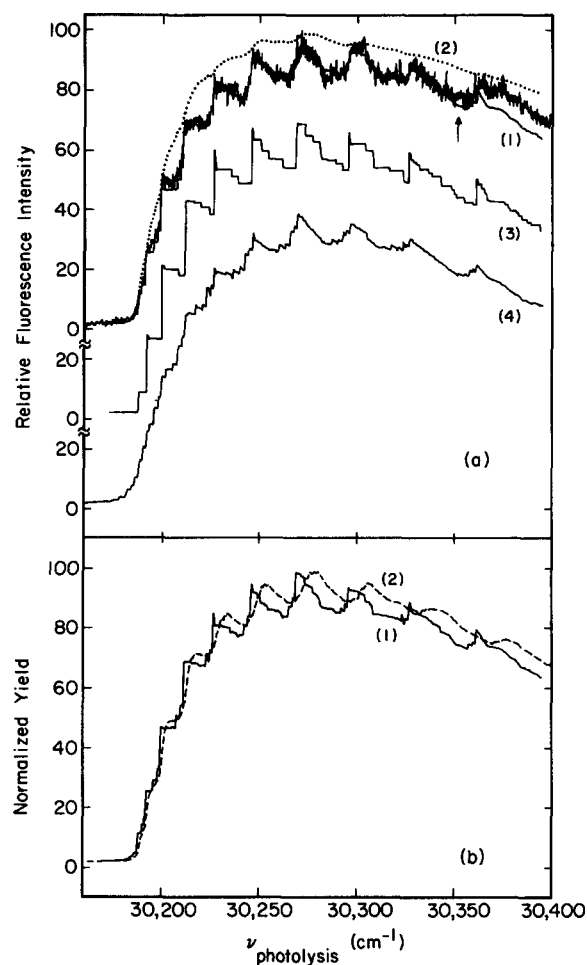


FIG. 4. (a) The effects of angular momentum conservation and the rotational temperature of ketene on the appearance of PHOFEX spectra. Shown are experimental data for the 2_{11} state [probe: $(0,14,0)3_{03} \leftarrow (0,0,0)2_{11}$ ($\tilde{b}-\tilde{a}$) transition at $16\,963.27\text{ cm}^{-1}$; nozzle diameter 0.5 mm] along with (1) the PST best fit ($T = 3\text{ K}$, $W'_i = 40$, $C_6 = \infty$, $b_{\text{max}} = \infty$), (2) the Prior theory fit (dotted) and PST curves for (3) $T = 0\text{ K}$, and (4) $T = 8\text{ K}$. The arrow indicates the threshold of a weak unassigned line overlapping the probed transition. (b) The effect of b_{max} on the PST fit for the 2_{11} state: (1) $b_{\text{max}} = \infty$ (solid), (2) $b_{\text{max}} = 4\text{ Å}$ (dashed).

pulses for the 0_{00} and 1_{01} states of singlet methylene with the photolysis energy fixed at less than 10 cm^{-1} above their respective thresholds. The signal rises to its maximum within the experimental time resolution of about 15 ns (Fig. 7). This implies that dissociation to form triplet methylene at $30\,120\text{ cm}^{-1}$ occurs at a rate greater than $7 \times 10^7\text{ s}^{-1}$.

As a PHOFEX spectrum is scanned from below threshold the noise increases markedly with the signal intensity above the background due to the long-lifetime fluorescence noted by Monts *et al.*³⁶ The shot-to-shot noise contains con-

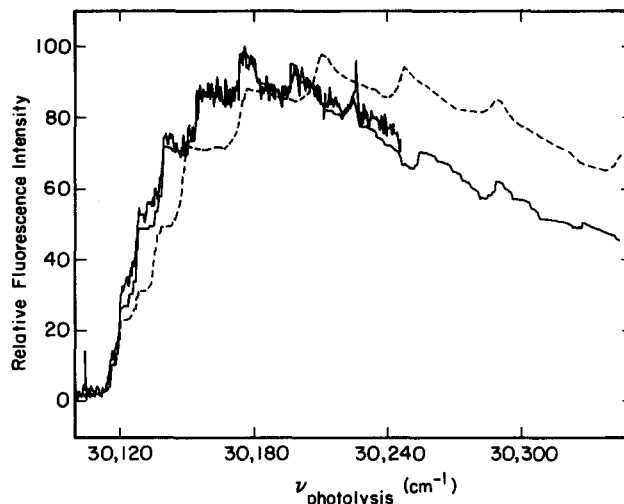


FIG. 5. Comparison of SACM (dashed, $\alpha/\beta = 0.5$) with PST (solid, $W'_i = 26$, $T = 3\text{ K}$, $C_6 = \infty$, $b_{\text{max}} = \infty$) and experiment for the 0_{00} state, probing the $(0,15,0)1_{10} \leftarrow (0,0,0)0_{00}$ ($\tilde{b}-\tilde{a}$) transition at $17\,742.65\text{ cm}^{-1}$, with a nozzle diameter of 0.5 mm . The very sharp peak just below threshold is due to a UV multiphoton resonance.

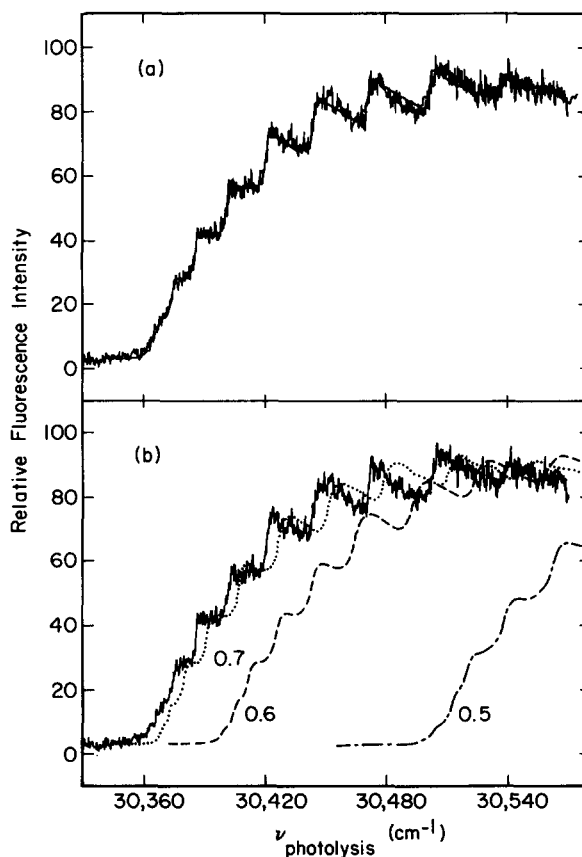


FIG. 6. (a) Experimental data and PST fit ($W'_i = 40$, $T = 3\text{ K}$, $C_6 = \infty$, $b_{\text{max}} = \infty$) for the 5_{15} state. Experimental data measured probing the $(0,14,0)5_{05} \leftarrow (0,0,0)5_{15}$ ($\tilde{b}-\tilde{a}$) transition at $16\,932.30\text{ cm}^{-1}$, with a nozzle diameter of 0.5 mm . (b) Effects of changing (α/β) in the SACM model. The experimental data from (a) is plotted (solid) with SACM fits using $\alpha/\beta = 0.5$ (---), $\alpha/\beta = 0.6$ (dashed), and $\alpha/\beta = 0.7$ (dotted). SACM is virtually indistinguishable from PST for $\alpha/\beta > 0.75$.

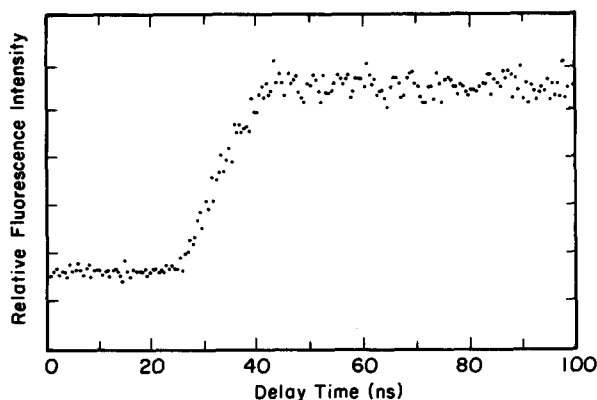


FIG. 7. The rise time of $^1\text{CH}_2$ from ketene photolysis at 332 nm, probing the $(0,15,0)1_{10} \leftarrow (0,0,0)0_{00}$ ($\tilde{b} \leftarrow \tilde{a}$) transition at $17\,742.65\text{ cm}^{-1}$. The x axis indicates the delay between the photolysis and probe pulses, plus an offset. The rise time measured here, close to the singlet channel threshold, is identical to that measured at 330 nm indicating that it is limited by the experimental time resolution.

tributions from fluctuations in the photolysis and probe laser powers, shot noise of the fluorescence photons, and variation in the number density of ketene in the probed region. Sources of drift and systematic error in the spectra are discussed in Sec. V F.

IV. DYNAMICAL MODELS

A. Conserved quantities

Certain quantum numbers are conserved throughout dissociation.^{44,45} In the absence of external fields, total energy (E) and total angular momentum including nuclear spin (F, M_F) are conserved. Neglecting the extremely small effects of the weak interaction, parity with respect to inversion (Π) is also conserved. Parity conservation restricts the orbital angular momentum quantum number l to be either even or odd, depending on the symmetry of the initial rovibronic state of the dissociating molecule.⁴⁴ In the case of ketene excited from its vibronic ground state to form $^1\text{CH}_2$, the conservation rule is

$$1 + J_{\text{CO}} + K_c''(\text{CH}_2\text{CO}) + K_c(\text{CH}_2) = \text{odd}. \quad (1)$$

Because the even and odd K_c'' states of ketene are almost equally populated and the number of even and odd l states of the products are nearly identical, this restriction has very little effect on the appearance of the PHOFEX spectra under the experimental conditions employed. Although this conservation law is not tested by the experimental data presented here, it was included in the fitting models for completeness.

Nuclear spin (I, M_I) is also expected to be conserved on the short time scale of dissociation because of the small size of the couplings between the nuclear spin and the rovibronic coordinates.^{44,45} If it is conserved, then J and M_J are also good quantum numbers ($F = J + I$). Conservation of nuclear spin has been experimentally tested and verified for the

photodissociation of formaldehyde to hydrogen plus carbon monoxide.⁴⁶

The difference in the thresholds for *ortho* and *para* singlet methylene states shown in Table I demonstrates that nuclear spin is conserved in the dissociation of ketene. Along with the threshold energies are listed the threshold differences between the various $^1\text{CH}_2$ states and the 0_{00} state, subtracting the $^1\text{CH}_2$ term values known from high resolution spectroscopy.^{47,48} The thresholds of *ortho* CH_2 (nuclear spin $I = 1$; $K_a + K_c = \text{odd}$) states are consistently about 9.7 cm^{-1} lower than expected from their term values. This can be understood only if nuclear spin is conserved throughout the photodissociation: the lowest $I = 1$ state of ketene ($J = 1$, $K_a = 1$) lies 9.7 cm^{-1} higher than the lowest $I = 0$ state ($J = 0$).⁴⁹ Although the accurate thresholds shown in Table I are obtained by fitting the PHOFEX curves using the PST model described below, the 9.7 cm^{-1} threshold displacement of the *ortho* states is so large that it is obvious even without benefit of the model.

Because no features are observed $5\text{--}10\text{ cm}^{-1}$ below the expected thresholds in the PHOFEX spectra of $I = 0$ methylene states⁴³ (e.g., Figs. 4–6), it can be determined that less than 5% of the observed $I = 0$ $^1\text{CH}_2$ originates from $I = 1$ ketene. In the cold free-jet expansion $kT_{\text{rot}} < 3\text{ cm}^{-1}$, so only the very lowest rotational states of ketene are expected to have appreciable populations. The roughly 3:1 intensity ratio of $I = 1$: $I = 0$ lines in the methylene LIF spectra measured in these experiments indicates that the nuclear spin of ketene is not significantly cooled in the supersonic expansion. This has often been observed in molecular beam spectra.^{50–52} An upper limit on the probability of spin conversion

TABLE I. Thresholds for production of CH_2 ($\tilde{a}^1A_1$) rotational states from CH_2CO (cm^{-1}). The thresholds indicate that nuclear spin is conserved in the photodissociation.

$^1\text{CH}_2 J_{K_a K_c}$	E_{th}^a	$E_{\text{rot}}^b + D_0^c$	$E_{\text{th}} - (E_{\text{rot}} + D_0)$
<i>para</i>			
0_{00}	30 116.2	30 116.2	0.0
1_{11}	30 143.4	30 143.4	0.0
2_{02}	30 169.8	30 169.8	0.0
2_{11}	30 187.9	30 188.1	− 0.2
4_{13}	30 326.1	30 326.3	− 0.2
5_{15}	30 364.6	30 364.1	+ 0.5
<i>ortho</i>			
1_{01}	30 125.0	30 134.5	− 9.5 ^d
1_{10}	30 137.6	30 147.5	− 9.9
2_{12}	30 166.0	30 175.8	− 9.8
3_{03}	30 210.9	30 220.5	− 9.6
3_{12}	30 238.4	30 248.2	− 9.8
5_{14}	30 411.0	30 420.6	− 9.6
6_{16}	30 446.5	30 455.9	− 9.4

^a Threshold energies extracted using the PST fitting model (see the text). The error bars are estimated to be $\pm 0.5\text{ cm}^{-1}$.

^b $^1\text{CH}_2$ rotational term values from Ref. 48.

^c The threshold for singlet dissociation $D_0 = E_{\text{th}}(0_{00}) = 30\,116.2 \pm 0.4\text{ cm}^{-1}$.

^d The lowest *ortho* state of CH_2CO is 1_{11} ; $E(1_{11}) = 9.7\text{ cm}^{-1}$.

$I = 1 \rightarrow 0$ during dissociation may therefore be set at 1.7%.

B. Physical models for PHOFEX spectra

The PHOFEX signal is proportional to the number of $^1\text{CH}_2$ molecules produced in the $J_{K_a K_c}$ state probed. This can be expressed as a sum over all the initial $J''_{K_a K_c}$ states of ketene (the background from vibrationally excited ketene is negligible):

$$S \propto \sum_{J''_{K_a K_c}} N(J''_{K_a K_c}) P(J_{K_a K_c}; J''_{K_a K_c}; h\nu), \quad (2)$$

where N is the number of ketene molecules in the specified rotational state, and P represents the probability that a particular $J''_{K_a K_c}$ state of ketene will be converted into the probed $J_{K_a K_c}$ state of methylene at the photolysis frequency ν . The threshold data, Sec. IV A, show that nuclear spin is conserved through the photodissociation, i.e.,

$$P(J_{K_a K_c}; J''_{K_a K_c}; h\nu) = 0$$

unless

$$K_a + K_c + K''_a = \text{even}.$$

The population distribution of ortho or para ketene is taken to be Boltzmannian with a rotational temperature T :

$$N(J''_{K_a K_c}) \propto (2J'' + 1) \exp(-E''_{\text{rot}}/kT), \quad (3)$$

$$E''_{\text{rot}} = (A - \bar{B})(K''_a)^2 + \bar{B}J''(J'' + 1), \quad (4)$$

$$\bar{B} = (B + C)/2, \quad (5)$$

using the symmetric top approximation for ketene ($\kappa = -0.997$).⁴⁹ The $I = 1$ and $I = 0$ molecules are completely separated in the calculations presented here, so no consideration of nuclear spin distributions is necessary.

The dissociation distribution function P can be written as the product of a dipole transition probability to an excited ketene state $J'_{K'_a K'_c}$ (the vibronic quantum numbers are suppressed) and a quantum yield for dissociation from that state to a particular $^1\text{CH}_2$ state $J_{K_a K_c}$:

$$P(J_{K_a K_c}; J''_{K_a K_c}; h\nu) = \sum_{J'_{K'_a K'_c}} \sigma(J'_{K'_a K'_c} \leftarrow J''_{K_a K_c}; h\nu) \Phi(J_{K_a K_c}; J'_{K'_a K'_c}; E), \quad (6)$$

where

$$E = h\nu + E''_{\text{rot}}. \quad (7)$$

Making the usual separation of rotational and vibronic coordinates, the dipole transition cross section can be written

$$\sigma(J'_{K'_a K'_c} \leftarrow J''_{K_a K_c}; h\nu) = \sigma_{\text{vib}}(h\nu + E''_{\text{rot}} - E'_{\text{rot}}) \sigma_{\text{rot}}(J'_{K'_a K'_c} \leftarrow J''_{K_a K_c}). \quad (8)$$

The angular momentum and parity change in a well-known manner upon optical excitation, while the nuclear spin quantum numbers are conserved.^{53,54} The absorption spectrum of the $\tilde{A}-\tilde{X}$ transition of ketene is known to be diffuse,⁴¹ even at -78°C ,⁴² presumably because of strong coupling between the $\tilde{A}$ state and the two lower-energy electronic states lead-

ing to rapid dissociation. In the absence of high resolution information on the ketene $\tilde{A}-\tilde{X}$ transition, it is assumed that the vibronic absorption cross section is smooth and slowly varying at the $\approx 0.3 \text{ cm}^{-1}$ laser resolution

$$\sigma_{\text{vib}}(h\nu + E''_{\text{rot}} - E'_{\text{rot}}) = \sigma_{\text{vib}}(h\nu). \quad (9)$$

$\sigma_{\text{vib}}(h\nu)$ is treated as essentially constant over short ($\approx 300 \text{ cm}^{-1}$) PHOFEX scans, although for longer scans a slow variation of the absorption cross section with laser frequency has been observed.¹⁵ The room-temperature absorption spectrum of ketene exhibits features no larger than $\pm 5\%$ in the region of interest,⁴¹ and the 10 K PHOFEX $\pm 15\%$.¹⁵ The rotational line strengths were approximated with symmetric top Hönl-London factors⁵³:

$$\sigma_{\text{rot}}(J'_{K'_a K'_c} \leftarrow J''_{K_a K_c}) = A_{KJ}(J'_{K'_a} \leftarrow J''_{K_a}). \quad (10)$$

The simulated PHOFEX curves were found to be insensitive to whether the Hönl-London factors for parallel or perpendicular transitions were employed. A change in the angular momentum quantum numbers of one unit more or less at constant energy is not critical in the statistical theories employed. Perpendicular transition line strengths were used in the fits presented here.

In statistical theories, the dissociation quantum yield Φ is expected to depend only on the energy and the conserved quantum numbers. Hence the quantum yield, Eq. (6), may be rewritten

$$\Phi(J_{K_a K_c}; J'_{K'_a K'_c}; E) = \Phi(J_{K_a K_c}; J'; \Pi; E), \quad (11)$$

where J' , Π , and E are conserved through the dissociation. Within the symmetric top approximation employed, K'_c is significant only in that it determines whether the parity with respect to inversion Π is even or odd. As mentioned above, the nuclear spin I is also considered to be conserved throughout.

The quantum yield reflects a competition between the particular $J_{K_a K_c}$ state of methylene probed and all the other accessible singlet and triplet methylene states:

$$\Phi = \frac{k_s(J_{K_a K_c})}{k_t + \sum_{J_{K_a K_c}} k_s(J_{K_a K_c})}, \quad (12)$$

where k_s and k_t represent the rate constants for dissociation into $^1\text{CH}_2$ and $^3\text{CH}_2$. Within statistical theories²⁻⁶ the unimolecular dissociation rate constant is given by $k = W/h\rho$ (strictly, $k \leq W/h\rho$),⁴ where W is the number of open channels and ρ is the reactant density of states. Hence

$$\Phi = \frac{W(J_{K_a K_c}; J'; \Pi; E)}{h\rho k_t + \sum_{J_{K_a K_c}} W(J_{K_a K_c}; J'; \Pi; E)}. \quad (13)$$

Over the $\approx 300 \text{ cm}^{-1}$ range of a single PHOFEX scan, which must lie at least 1625 cm^{-1} above the triplet barrier, the triplet rate constant k_t is considered to be invariant, and to be the same for all J' values. ρ is also considered to be slowly varying with energy, but $\rho \approx \rho_{\text{vib}}(2J' + 1)$.⁴ Hence

$$h\rho k_t = W'_t(2J' + 1), \quad (14)$$

where W'_t is varied to fit the data. W'_t can be interpreted as the effective number of triplet channels for $J' = 0$. It should be noted that the product distribution may be predicted cor-

rectly by the statistical models even when the absolute rate constant is not.^{6,18,19,43} W is calculated using different statistical theories.

1. Phase space theory

In the phase space theory,^{2,3}

$$W = \sum_{J_{\text{CO}}=0}^{J_{\text{CO}}^{\text{max}}} \sum_{jj=|J_{\text{CO}}-J|}^{J_{\text{CO}}+J} \sum_{\substack{l=\lceil J'-jj \rceil \\ \text{even/odd}}}^{\min(l_{\text{max}}, J'+jj)} 1, \quad (15)$$

where the maximum J_{CO} is set by energy conservation and the even/odd condition is set by parity conservation, Eq. (1). The quantum number jj quantizes the vector sum $\mathbf{J} + \mathbf{J}_{\text{CO}}$, and the bounds involving jj are set by angular momentum conservation.³ Additional constraints beyond the conservation laws are incorporated into PST through l_{max} . One common constraint is the inclusion of a centrifugal barrier. A long-range attractive potential of the form $V = -C_6 R^{-6}$ is assumed, and this constraint condition becomes

$$\hbar^2 l_{\text{max}} (l_{\text{max}} + 1) < 3\mu (2C_6)^{1/3} E_{\text{trans}}^{2/3}, \quad (16)$$

where μ is the reduced mass of CH_2 and CO. The translational energy of the products

$$E_{\text{trans}} = E - (D_0 + E_{\text{CH}_2} + E_{\text{CO}}). \quad (17)$$

D_0 is the threshold energy for singlet dissociation, and the CH_2 and CO rotational term values are taken from high resolution spectroscopic data.^{47,48,55} Another reasonable constraint is to limit the maximum size of the impact parameter b between the separating fragments. Then

$$\hbar^2 l_{\text{max}} (l_{\text{max}} + 1) < 2\mu E_{\text{trans}} b_{\text{max}}^2. \quad (18)$$

Both of these constraints, Eqs. (16) and (18), were included in the fitting program, although they can be effectively removed by setting C_6 and b_{max} large enough. (The experimental data do not exhibit the effects of these constraints because angular momentum conservation provides a much more severe bound on l at the small J values accessed here. Hence only lower bounds on C_6 and b_{max} are obtained from the fits, Sec. V B.)

A variation on PST, the Prior calculation of Levine and Kinsey,¹ ignores angular momentum conservation by replacing the summation over l with a term proportional to $(E_{\text{trans}})^{1/2}$. No additional constraints such as centrifugal barriers are included in the Prior theory.

The energy resolution of most of the computations is 0.5 cm^{-1} , but in the threshold region finer-grained calculations with 0.1 cm^{-1} resolution were also performed.

2. Statistical adiabatic channel model (SACM)

In SACM, the states are counted in much the same way as in PST, but effective barriers are introduced for each different product channel in addition to the centrifugal barrier, causing a further contraction of the available phase space. Algebraically,

$$W = \sum_{J_{\text{CO}}=0}^{J_{\text{CO}}^{\text{max}}} \sum_{jj=|J_{\text{CO}}-J|}^{J_{\text{CO}}+J} \sum_{l=\lceil J'-jj \rceil}^{J'+jj} h(x), \quad (19)$$

$$x = E - V_{\text{max}}(J_{\text{CO}}, J_{K_a K_c}, J', l, jj),$$

where $h(x) = 1$ if $x \geq 0$ and 0 otherwise, and the symbols have the same meaning as in Eq. (15). For simplicity, parity conservation is neglected in the SACM calculations, as has been done previously.⁴⁻⁷ The effective barrier height V_{max} for each dissociation channel is given by the maximum of the effective SACM potential $V_{\text{eff}}(q)$ for $q > 0$, where the displacement along the reaction coordinate $q = R_{\text{c.m.}} - R_e$. $R_{\text{c.m.}}$ represents the separation of the centers of mass of CO and CH_2 ; R_e is the equilibrium separation in ketene. V_{eff} for the i th product channel is given by⁴⁻⁷

$$V_{\text{eff}}(q) = V_M(q) + \Delta E_{\text{zp}} e^{-\alpha q} + E_{\text{ketene}}(i) e^{-\alpha q} + \bar{B}(q)P(P+1) + E_{\text{prod}}(i)(1 - e^{-\alpha q}), \quad (20)$$

where

$$P = 1 + (J' - 1)e^{-\alpha q} \quad (21)$$

and the potential energy takes the form of a Morse potential

$$V_M(q) = (D_0 + \Delta E_{\text{zp}})(1 - e^{-\beta q})^2. \quad (22)$$

ΔE_{zp} is the zero-point energy difference between ketene and the products. α and β are treated as parameters to fit the data, within a limited range where they are physically reasonable.⁴ Dissociation is taken to occur along the C–C bond direction so $\bar{B}(q) = \bar{B}R_e^2/(R_e + q)^2$. The spectroscopic term values are used to determine the product internal energy

$$E_{\text{prod}} = E_{\text{CH}_2}(J_{K_a K_c}) + E_{\text{CO}}(J_{\text{CO}}). \quad (23)$$

The harmonic-oscillator symmetric-top approximation is used for the ketene energies [the term $\bar{B}J'(J'+1)$ is treated separately]

$$E_{\text{ketene}} = E_{\text{vib}} + (A - \bar{B})K^2, \quad (24)$$

$$E_{\text{vib}} = \omega_5 \nu_5 + \omega_6 \nu_6 + \omega_8 \nu_8 + \omega_9 \nu_9. \quad (25)$$

ν_5 , ν_6 , ν_8 , and ν_9 are the “disappearing oscillator” modes which become the product rotations and translations. ν_4 corresponds to the reaction coordinate. In calculating $V_{\text{eff}}(q)$ the SACM assumes that other vibrational quantum numbers $(\nu_1, \nu_2, \nu_3, \nu_7)$, which correlate to the vibrations of the product CO and CH_2 , are individually conserved.^{4,6,26} Here only vibrational ground state products are energetically accessible, so only the $(\nu_1, \nu_2, \nu_3, \nu_7) = 0$ channels can be open.

In the calculation, the different $E_{\text{ketene}}(\nu_5, \nu_6, \nu_8, \nu_9, K)$ are generated and sorted in order of increasing energy. Likewise all the product states $|J_{K_a K_c}, J_{\text{CO}}, l, jj\rangle$ are listed and labeled with an energy index “ i .” Degenerate product states are indexed in order of increasing l , or, if they have the same l value, increasing jj .^{4,56} In generating V_{eff} , E_{ketene} states of the same index as the product state of interest are employed. Increased effective barriers for higher energy product states result from this process of adiabatically correlating product and reactant states. The indexing procedure must be repeated for each different value of J' .

The maximum of V_{eff} is found by a grid search on q , with a grid spacing of $0.05 \text{ \AA}$. Because of the multiple nested summations and the indexing procedure the SACM program requires a fair amount of computer time, and the time re-

quired increases rapidly with photolysis energy. To reduce the computational difficulty Troe and co-workers have devised various approximation techniques for SACM.^{7,23,24} One of these, the SACM-PST model, is discussed below.

3. The SACM-PST model

In SACM-PST⁷ effective rotational constants for the product states rather than the spectroscopic ones are used to calculate the PST W in Eq. (15). In deriving the effective rotational constants, each product rotation is correlated with one of the “disappearing oscillators”²³

$$B_{\text{eff}} = B_{\text{true}} + (\epsilon - B_{\text{true}}) \exp[-C_3(\alpha/\beta) - C_4(\alpha/\beta)^4], \quad (26)$$

$$C_3 = 4.00 - 1.29 \ln(\epsilon/D_0), \quad (27)$$

$$C_4 = 2.80 - 5.19 \ln(\epsilon/D_0), \quad (28)$$

where D_0 is the dissociation threshold and ϵ is the vibrational frequency of the disappearing oscillator correlated with the rotation. The B constant of $^1\text{CH}_2$, which correlates with the K_a rotor of ketene, is basically unchanged by this conversion to effective constants. The in-plane and out-of-plane CH_2 wags in ketene correlate to the C and A constants of $^1\text{CH}_2$, respectively, and the two CCO bends correlate with B of CO. As was done by Brouwer *et al.*,⁷ the geometric mean of the effective B_{CO} values was used in the SACM-PST simulations. For the “typical” value of $\alpha/\beta = 0.5$, the effective constants C , A , and B_{CO} exceed the spectroscopic values by factors of 1.5, 1.1, and 1.5, respectively. SACM-PST was compared with the exact SACM calculations, and found to be a rather poor approximation in the threshold region,⁴³ in contrast with previous results at higher energies.⁷

V. DYNAMICAL MODEL FITS OF PHOFEX SPECTRA

The phase space theory model is found to provide excellent fits to the experimental data, Figs. 2–6 and Ref. 43. The base lines and the heights of the calculated PHOFEX spectra are scaled to match the experimental data. The singlet channel dissociation threshold D_0 is accurately determined from the fits. The free parameters in the PST model are discussed in order of their influence on the fit. The results of the Prior, SACM, and SACM-PST fitting models are also briefly outlined. Finally, the deviations of the PST fits from the experimental data are critically examined.

A. Dissociation threshold measurements

Table I lists the experimental thresholds of the methylene ($\tilde{a} \ ^1A_1$) rotational states for which PHOFEX spectra have been obtained in the present work. Because very cold ketene was photolyzed with narrow bandwidth ultraviolet light at the photochemical threshold, these measurements provide the most accurate value yet obtained for the dissociation energy, and thus the heat of formation of the methylene radical. The PHOFEX threshold of an unassigned $\tilde{b} - \tilde{a}$ LIF spectral line gives the $\tilde{a}$ -state rovibrational term value. In this way many previously unreported LIF transitions of this highly perturbed spectrum^{47,48} were observed and as-

signed. These spectroscopic results will be reported separately.⁵⁷

The thresholds given in Table I were extracted from the PST fits to the experimental data. The thresholds are precisely determined by the data, as illustrated in Fig. 3. The error bars are found from comparisons of this type on many different scans. In the fitting models it is assumed that nuclear spin is conserved; relaxing this restriction would cause all the calculated PHOFEX curves for the *para* states to be offset about 9.7 cm^{-1} from the experimental curves. The differences between the thresholds match the $^1\text{CH}_2$ term values; no barriers which differ for different rotational states are detected. Combining the data for the various $^1\text{CH}_2$ states probed yields a singlet dissociation threshold D_0 of $30\,116.2 \pm 0.4 \text{ cm}^{-1}$.

The dissociation threshold value obtained agrees well with previous measurements. The earlier PHOFEX study, Bitto *et al.*,¹⁵ reported the threshold to be $30\,102 \pm 15 \text{ cm}^{-1}$. The accuracy was limited by the thermal energy of ketene. In an earlier molecular beam study using fixed-frequency photolysis lasers Hayden *et al.*³¹ measured the value $30\,100 \pm 170 \text{ cm}^{-1}$.

Combining the threshold value with other data yields additional thermochemical results. Using spectroscopic data for methylene, Bunker *et al.*²⁷ found the singlet–triplet splitting to be $\Delta_{ST} = 3156 \pm 5 \text{ cm}^{-1}$. Hence the dissociation energy of ketene $D(\text{CH}_2\text{--CO}) = 26\,960 \pm 6 \text{ cm}^{-1}$. Combining this result with the heat of formation of ketene, $-10.7 \pm 0.4 \text{ kcal/mol}$, measured by Nutall *et al.*⁵⁸ yields $\Delta H_{f,0}^\circ(\text{CH}_2) = 93.6 \pm 0.4 \text{ kcal/mol}$. These results are in excellent agreement with the results of McCulloh and Dibeler⁵⁹ obtained from dissociative photoionization of ketene and methane: $D(\text{CH}_2\text{--CO}) = 26\,900 \pm 70 \text{ cm}^{-1}$; $\Delta H_{f,0}^\circ(\text{CH}_2) = 93.6 \pm 0.4 \text{ kcal/mol}$.

B. Effects of the model parameters on the fits

The rate of dissociation on the triplet surface, or more precisely the effective number of triplet channels W'_t [Eq. (14)], is an important parameter in the fits. Increasing W'_t shifts the maximum of the calculated PHOFEX curve to higher energy. As illustrated in Fig. 2(b), the fits are relatively sensitive to the value of W'_t employed and each PHOFEX curve defines a value for W'_t to within 25% (though systematic errors may increase this uncertainty to $\approx 50\%$, Sec. V F). Table II lists the best fitting values of W'_t and the corresponding triplet channel dissociation rates. The parameter W'_t also measures how rapidly the singlet/triplet branching ratio increases with photolysis energy, as discussed in Sec. VI below. The rate values listed in Table II are reasonably close to those estimated in Sec. VI by other experimental and theoretical techniques. However, these rate estimates assume that PST accurately gives the singlet channel rate, as discussed in Ref. 43.

In all the fits shown here, the rotational temperature of ketene is set at 3 K. The choice of rotational temperature is consistent with the results of the D_2CO LIF measurement. Higher temperatures (e.g., $T = 8 \text{ K}$) begin to noticeably smooth out the fine structure in the calculated PHOFEX

curves, Fig. 4(a). The quality of the fits near threshold, Fig. 3, is a further indication that $T \approx 3$ K.

The fits are not very sensitive to C_6 and $b_{\max}$. The Lennard-Jones parameter C_6 is found to be constrained by the data to be greater than $5 \times 10^4 \text{ cm}^{-1} \text{ \AA}^6$. (This is less than C_6 values quoted for the Ne-Ne interaction.⁶⁰) Smaller values of C_6 significantly smooth the steps on the calculated PHOFEX curves. However, the fit is not sensitive to larger values of C_6 . Physically, $V = -C_6 R^{-6}$ is the attractive potential which lowers the centrifugal barrier, and the parameter C_6 sets the height of that barrier. If there were a significant centrifugal barrier, each orbital angular momentum value l would have a slightly different appearance threshold, because the higher l states would have higher centrifugal barriers. The overall effect would be to make the steps in the PHOFEX curves rise more gradually with energy; if the barrier is sufficiently high, the steps are altogether smoothed out. The data require that there be no centrifugal barrier greater than 5 cm^{-1} at least up to $l = 10$. (Put another way, the centrifugal barriers must all occur at C-C bond lengths greater than about $3 \text{ \AA}$.) The form of the attractive potential is not critical, as long as it is sufficiently attractive to effectively erase the centrifugal barrier. In all of the fits shown here $C_6 = \infty$, i.e., there are no centrifugal barriers.

The maximum allowed impact parameter $b_{\max}$ is constrained by the data to be greater than $5 \text{ \AA}$ for similar reasons. Smaller values of $b_{\max}$ restrict the possible values of l at a specified energy. This particularly limits the number of channels opening just above an energy threshold. As the energy increases, the higher l values gradually become accessible. The effect of smaller $b_{\max}$ is to smooth out the features in the calculated PHOFEX curves, Fig. 4(b). In all of the fits to the data shown here there is no restriction on the size of the impact parameter.

The experimental thresholds and the sharp steps in the PHOFEX spectra indicate unambiguously that there are essentially no barriers to dissociation on any allowed channel. The excellent fits to the experimental data provided by PST indicate that all the product channels allowed by the conservation laws discussed in Sec. IV A are open.

C. The Prior model

The Prior model, which is a version of phase space theory without inclusion of angular momentum conservation, does not reproduce the sharp steps in the PHOFEX spectra, Fig. 4(a). The sharpness of the fine structure in the PHOFEX curves calculated using the PST model is due to quick opening of a number of product l, j channels each time an additional CO rotational level becomes energetically accessible. The number of l, j channels increases with total energy until they reach the constraints imposed by angular momentum conservation, where there is a sharp cutoff. In the Prior calculation the orbital angular momentum l increases beyond this cutoff, yielding a gradual rise to the next CO channel.

D. The statistical adiabatic channel model

The statistical adiabatic channel model becomes equivalent to phase space theory as $\alpha \rightarrow \infty$. As pointed out by

Troe,²⁰ the SACM actually converges to PST when $\alpha/\beta \rightarrow 1$, and the present data cannot distinguish between PST and SACM for $\alpha/\beta > 0.75$. In this limit, the SACM results provide excellent fits to the data. However, for the suggested "standard" value²³ of $\alpha/\beta = 0.5$ the SACM calculation fails to correctly reproduce the fine structure in the experimental PHOFEX curves (Figs. 5 and 6). Also, SACM increases the energy thresholds of the different $^1\text{CH}_2$ rotational states. This effect is particularly apparent at the higher rotational levels, as illustrated in Fig. 6(b). With the standard value of α/β , SACM fails because it introduces effective "barriers" which increase as additional channels open. The adiabatic barriers also reduce the attractive potential significantly at large distances, causing the centrifugal barriers to occur at smaller impact parameters. This significantly smooths the fine structure in the PHOFEX spectra, because different l states now have significantly different thresholds. In the fits shown in Figs. 5 and 6 the Morse parameter β is fixed at $2.3 \text{ \AA}^{-1}$, consistent with the vibrational frequency of the C-C stretch. The fits were found to be relatively insensitive to β at a fixed α/β ratio.

E. The SACM-PST model

As has been noted before,⁷ SACM-PST is virtually identical to PST for values of $(\alpha/\beta) > 0.7$. For smaller values of α/β the SACM-PST approximation still produces structured PHOFEX curves; however, the spacings between the sharp features become larger than the experimental spacings (which match the CO term values), and, like SACM, the $^1\text{CH}_2$ thresholds are also greater than those found by experiment.⁴³ SACM-PST does not match detailed SACM very well in calculations so close to threshold, in contrast to the much better agreement found at higher energies for H_2O_2 .^{7,43}

F. Deviations from the PST fits

Because of the excellent quality of the phase space theory fits, it is interesting to examine the relatively small deviations of the fits from the PHOFEX data. Some of these deviations are attributable to finite signal to noise, but the most troublesome problems in the PHOFEX data are due to systematic errors. The greatest difficulties are with the triplet dissociation rate parameter W'_i , which is well defined by the fit for each state but varies erratically from state to state (Table II), suggesting the presence of a systematic error. Any effect which adds a long-term drift or slope to the experimental data will affect the value of W'_i extracted in the fitting procedure. The uncertainty in W'_i limits the degree to which the data can demonstrate that there are no important dynamical constraints in the singlet dissociation in the energy region probed. Some possible sources of this error are discussed as follows.

(1) *Overlaps in the $^1\text{CH}_2$ LIF spectrum:* Some of the observed deviations between the data and the fits are due to overlaps between the probed transitions and other weak ($< 15\%$ of the intensity of the strong transition), unassigned transitions which grow in at higher photolysis energy.

TABLE II. The effective number of triplet channels W'_i and the corresponding triplet channel dissociation rates.

$^1\text{CH}_2$ state	W'_i ^a	k (s ⁻¹) ^b
0 ₀₀	26(4)	1.6×10^8
1 ₀₁	40(5)	2.4×10^8
1 ₁₁	24(6)	1.4×10^8
1 ₁₀ ^c	31(5)	1.8×10^8
2 ₀₂	30(5)	1.8×10^8
2 ₁₂	31(10)	1.8×10^8
2 ₁₁ ^c	40(5)	2.4×10^8
3 ₀₃	53(10)	3.1×10^8
3 ₁₂	120(20)	7.0×10^8
4 ₁₃ ^d	50(15)	2.9×10^8
5 ₁₅	40(5)	2.3×10^8
5 ₁₄	110(30)	6.3×10^8
6 ₁₆ ^d	300(100)	1.7×10^9

^a From the PST fits to the PHOFEX data. See Eq. (14). Estimated experimental uncertainties in parentheses.

^b Triplet channel dissociation rate $k = W'_i/h\rho$, calculated using $\rho(\tilde{X})$ from the Whitten–Rabinovitch formula.

^c Transition overlapped by a weak unassigned transition which appears at higher photolysis energy. W'_i determined by fitting curve up to threshold of the interfering line.

^d Value may be unreliable, because transition is overlapped by a weak transition with a lower appearance threshold.

The presence of overlaps was detected by high resolution (Doppler-limited, 0.1 cm^{-1})⁴³ measurements of the LIF excitation spectrum. Overlaps were found for several transitions, Table II. This problem is more severe at higher photolysis energies where the $^1\text{CH}_2$ spectrum becomes quite congested.^{15,47,48} The overlaps are responsible for the largest deviations from the fits [Fig. 4(a)]; however, they do not explain the variability in the values of W'_i extracted.

(2) *Variations in the absorption of ketene*: Another possible source of systematic deviations between the PHOFEX data and the PST fits is that the absorption cross section of ketene varies with the photolysis laser frequency, rather than being wavelength independent as assumed in the model calculation. Attempts were made to correct for the features seen in the room-temperature absorption spectrum of ketene⁴¹ and in a previous PHOFEX study.¹⁵ An absorption spectrum of jet-cooled ketene would provide a more accurate way to make these corrections, but none is currently available. These corrections cannot explain the erratic variations in W'_i nor the observed deviations from the PST fits, since these do not occur at the same photolysis wavelength for scans of different singlet methylene states.

(3) *Multiphoton processes*: A general increase in the background level with photolysis laser frequency is also possible, the background being due to the multiphoton UV-induced luminescence as well as a variety of hypothetical mechanisms involving both visible and UV light. The background signal off-resonance was observed to be always less than 5% of the experimental signal, and so not able to explain the very large variations in W'_i . The PHOFEX curves were also observed to be independent of photolysis laser

power as it varied by more than a factor of 2, suggesting that two-photon processes are relatively unimportant compared to one-photon photolysis.

(4) *Experimental artifacts*: The most important source of systematic error contributing to the error in W'_i is the variation in the concentration of ketene in the carrier gas with time, largely due to the rise in the slush-bath temperature over the course of a half-hour scan. The vapor pressure of ketene is very sensitive to the temperature in the region of interest: a 1 K change in temperature would cause a 10% drift in the PHOFEX curve. To guard against this error, every scan was repeated scanning in the opposite direction. Only those data which were reproducible to better than 10%, corresponding to a 30% change in W'_i , were considered reliable.

Other experimental artifacts were also considered. Errors in the normalization procedure could potentially cause a significant drift of the observed PHOFEX curve. However, the same sorts of deviations between calculation and experiment are observed with the raw unnormalized data. The increasing Doppler width of methylene with increasing photolysis energy would be expected to slightly decrease the PHOFEX signal with energy, as noted by Bitto *et al.*, but the experimental PHOFEX intensity is generally greater than expected at high energy. Allowing W'_i to increase linearly with energy in the fits does not significantly improve the quality of the fits. Various saturation effects which could conceivably distort the PHOFEX spectra were tested by attenuating the laser beams. The spectra obtained, though of poorer quality than those reported here, are very similar in their gross features. They also exhibit the sharp fine structure, proving that it is not due to some nonlinear effect.

(5) *A failure of the PST?*: If the deviations between the model and the experimental PHOFEX data are not the result of some experimental artifact, one is forced to reconsider the assumptions in the model. One way to fit the deviations would be to introduce cleverly chosen dynamical constraints which restrict the accessible singlet product phase space by about 20%. (A 20% slope added to a PHOFEX curve would cause a much larger change in the value of W'_i extracted from the fit.) Another more appealing possibility is that the rate is not given correctly by PST except very near threshold, but the product state distribution is given by PST.⁴³ In any case, the data definitely rule out many kinds of barrier dynamical constraints which raise the appearance thresholds of different rotational states of $^1\text{CH}_2$, or which limit l as a function of the photolysis energy, yielding smoothed PHOFEX curves.

VI. DISSOCIATION RATES AND THE SINGLET/TRIPLET BRANCHING RATIO

Following excitation to $\tilde{A}^1A''$, the ketene molecule undergoes internal conversion to $\tilde{X}^1A_1$ and/or intersystem crossing to $\tilde{a}^3A''$ prior to unimolecular dissociation on the $\tilde{X}$ or $\tilde{a}$ potential surface, Fig. 1. Experiment provides some bounds on rate constants. The presence of some vibrational structure in the ketene $\tilde{A} \leftarrow \tilde{X}$ absorption spectrum⁴¹ with widths $\approx 200 \text{ cm}^{-1}$ may imply that the nonradiative decay rate of $\tilde{A}$ is less than $4 \times 10^{13} \text{ s}^{-1}$. The low fluorescence quan-

tum yield ($< 10^{-5}$), combined with the known oscillator strength, indicates that this rate is greater than $2.5 \times 10^9 \text{ s}^{-1}$.⁶² The rise time of the product $^1\text{CH}_2$ measured in the present work is less than 15 ns, even at threshold, indicating that the ketene photodissociation rate constant to yield triplet CH_2 at the singlet threshold is greater than $7 \times 10^7 \text{ s}^{-1}$. Several groups have photolyzed ketene at 313 nm and measured the yield of CO formed^{61,62}; their Stern–Vollmer analyses indicate a unimolecular dissociation rate constant on the order of 10^{10} s^{-1} at 1830 cm^{-1} above the singlet threshold.

A molecular beam study by Hayden *et al.*³¹ revealed no triplet product at 2350 cm^{-1} above the singlet threshold. No highly rotationally excited CO indicative of dissociation to $^3\text{CH}_2$ was found in a VUV LIF study at the same photolysis wavelength.¹⁶ From these results the branching ratio is certainly greater than 10:1 in favor of the singlet at $32\,466 \text{ cm}^{-1}$.

The PHOFEX spectra provide an indirect measurement of the singlet/triplet branching ratio near the singlet threshold. The branching ratio, given by the calculated number of singlet dissociation channels W_s divided by the effective number of triplet channels W'_t , increases from 0 to about 2 in the first 125 cm^{-1} above the singlet threshold. The sharp increase in the branching ratio with photolysis energy indicates that the singlet dissociation rate constant increases much more rapidly than the triplet rate constant. The accuracy with which the branching ratio can be determined is limited by the uncertainty in the fitting parameter W'_t , Sec. V F.⁴³

There are not enough experimental data to determine the mechanism by which the branching ratio increases so rapidly with energy. Two different physical models to explain this phenomenon are presented here, one in which the rate of formation of triplet is limited by the dissociation rate on the triplet surface; the second in which this rate is limited by the rate of intersystem crossing onto the triplet surface.

As a beginning, consider the density of states for each of the three electronic manifolds $\tilde{X}^1A_1$ (S_0), $\tilde{a}^3A'$ (T_1), and $\tilde{A}^1A''$ (S_1). The Whitten–Rabinovitch result⁶³ must be divided by 4 because there are four symmetry classes in the appropriate CNPI group⁵⁴; the effects of symmetry are included explicitly in the PST model presented in Sec. IV. For the ground state, the experimental vibrational frequencies⁶⁴ are employed, yielding $\rho(\tilde{X}) = 5000/\text{cm}^{-1}$ at the singlet channel threshold. This is expected to be an underestimate because the calculation does not correct for anharmonicity. The energies and vibrational frequencies of the other electronic states are not known experimentally, so these values are taken from the *ab initio* studies of Allen and Schaefer.^{28,29} For the $\tilde{a}$ state, the $\tilde{a}$ – $\tilde{X}$ energy gap T_0 is taken to be $19\,150 \text{ cm}^{-1}$ from a series of sophisticated CI calculations.²⁹ The scaled CISD $\tilde{a}$ state vibrational frequencies are also taken from these computations. No attempt was made to correct for the effects of the nearly degenerate $^3A'$ state which connects with the $^3A''$ state through a conical intersection.^{28,29} (This could conceivably increase densities of triplet states by up to a factor of 2.) The density of triplet states of

the proper symmetry $\rho(\tilde{a})$ is calculated to be $18/\text{cm}^{-1}$ at the singlet threshold. Less theoretical work has been done on the $\tilde{A}$ state of CH_2CO , but using the suggested $\tilde{A}$ – $\tilde{X}$ T_0 value²⁹ of $21\,100 \text{ cm}^{-1}$ and the scaled (0.9) SCF frequencies²⁸ (averaging the two theoretical values for ν_8 and ν_9), $\rho(\tilde{A}) = 5/\text{cm}^{-1}$ at the singlet threshold of $30\,116.2 \text{ cm}^{-1}$. The calculated densities of states increase rather slowly with energy in the singlet threshold region: increasing the energy by 100 cm^{-1} increases all the ρ 's by less than 6%. Using these densities of states it is possible to calculate the absolute rate constants for dissociation on the S_0 and T_1 surfaces from statistical theory. For the S_0 surface 100 cm^{-1} above threshold a dissociation rate constant k_d of $2 \times 10^8 \text{ s}^{-1}$ is calculated using PST.⁴³ The corresponding linewidth is $\approx 0.001 \text{ cm}^{-1}$, much larger than the average spacing between states of $1/\rho(\tilde{X}) = 0.0002 \text{ cm}^{-1}$. As discussed below, the T_1 surface dissociation rate constant k_T is calculated to be $\geq 4 \times 10^{10} \text{ s}^{-1}$, so the T_1 states also form a heavily overlapped continuum.

The PHOFEX spectra indicate that the quantum yield of $^1\text{CH}_2$ is proportional to the number of singlet product quantum states, at least near threshold. This shows that the $^1\text{CH}_2$ formation rate constant is limited by the dissociation rate on the S_0 surface k_d rather than by the rate of internal conversion from $S_1 \rightarrow S_0$; i.e., $k_{ic} \gg k_d$. Therefore, the internal conversion rate must be greater than 10^9 s^{-1} . To derive this rate using Fermi's Golden Rule, the corresponding rms coupling matrix elements between S_1 and S_0 must be greater than 0.0004 cm^{-1} .

The relatively slow 5 ns photolysis pulses excite a set of eigenstates of the form

$$|\Psi_n\rangle = \sum_i \alpha_{ni} |S_{1i}\rangle + \sum_j \beta_{nj} |S_{0j}\rangle + \sum_k \gamma_{nk} |T_{1k}\rangle, \quad (29)$$

where the S_0 and T_1 states are considerably lifetime broadened due to dissociation. It is expected that all the S_0 states of similar energy will have similar decay rates k_d and likewise all the T_1 states will decay at some nearly uniform rate k_T . Making this approximation, the dissociation rate constant of a particular ketene eigenstate is given by

$$k_n = k_d \sum_i |\beta_{ni}|^2 + k_T \sum_j |\gamma_{nj}|^2 \quad (30)$$

and the singlet–triplet branching ratio is given by

$$(\Phi_s/\Phi_t)_n = \frac{k_d \sum_i |\beta_{ni}|^2}{k_T \sum_j |\gamma_{nj}|^2}. \quad (31)$$

Eigenstates with the most S_1 character, i.e., the largest $\sum |\alpha_{ni}|^2$, will be preferentially excited by the laser, since only S_1 states carry significant oscillator strength.

A. The strong coupling limit ($k_{isc} \gg k_T$)

One interesting case to consider is the limit of strong coupling among the three electronic states compared with coupling to the dissociation continuum, i.e., internal conversion and intersystem crossing much faster than the dissociation rates on the T_1 and S_0 surfaces. In this limit,

$$\sum_i |\beta_{ni}|^2 \approx \rho(\tilde{X})/\rho_{\text{total}}, \quad \sum_j |\gamma_{nj}|^2 \approx \rho(\tilde{A})/\rho_{\text{total}}, \quad (32)$$

where ρ_{total} is the sum of the densities of states on the three electronic surfaces [and is dominated by $\rho(\tilde{X})$]. The summations will show some variation about these mean values,⁶⁵ but presumably these fluctuations will be relatively small. Combining Eqs. (30)–(32) and the relation $k = W/h\rho$ yields

$$k_n \approx (W_s + W_t)/h\rho_{\text{total}}, \\ (\Phi_s/\Phi_t)_n \approx W_s/W_t, \quad (33)$$

where W_s and W_t are the number of open reaction channels forming singlet and triplet methylene, respectively. These equations are essentially the same as those used to determine the fitting parameter W'_i in the PHOFEX fitting procedure, Sec. IV; if W_s is determined by PST, W_t should equal W'_i to match all the experimental data.

The triplet dissociation, which is known theoretically^{28,29} and experimentally³⁰ to proceed over a barrier, can be described by RRKM theory. The phase space theory (PST) rate may be used for the singlet channel. To explain the rapid increase in the branching ratio in this model the singlet channel transition state must be much "looser" than the triplet channel transition state, giving it a much larger " A factor." A similar explanation has been given for the rapid switching from molecular to radical products in the photodissociation of formaldehyde.^{23,66}

The number of open channels W_t and the dissociation rate on the $\text{CH}_2\text{CO } \tilde{a}^3A''$ surface k_T are estimated using RRKM theory.⁶⁴ The excess energy of the triplet channel transition state at the singlet threshold $E^\ddagger = \Delta_{ST} - E_b$ where Δ_{ST} is the singlet–triplet energy gap in methylene and E_b is the barrier height (Fig. 1). Δ_{ST} is known to be $3156 \pm 5 \text{ cm}^{-1}$ from spectroscopic data.²⁷ The barrier height has not been determined experimentally, although the high yield of carbon monoxide obtained with 351 nm photolysis,^{30,31} combined with Δ_{ST} and the singlet channel dissociation threshold D_0 determined in the present work, shows that it is less than 4.4 kcal/mol. The best *ab initio* value is 5.82 kcal/mol, but this value is expected to be an overestimate.²⁹ The transition state vibrational frequencies are taken from a scaled *ab initio* calculation at the CISD level.²⁹ No corrections for anharmonicity are applied. The torsion about the C–C bond " ν_9 " is actually a hindered rotor, with a barrier height of 420 cm^{-1} by *ab initio* calculation.²⁹ The hindered rotor eigenvalues were found by assuming a sinusoidal potential, and an effective moment of inertia appropriate to the rotation of the CH_2 group about its symmetry axis ($1.91 \text{ amu } \text{\AA}^2$), and solving the one-dimensional Schrödinger equation numerically. (This treatment of ν_9 gives a state count $\approx 50\%$ larger than the harmonic oscillator treatment.) The number of vibrational states of the transition state was found by direct count, and divided by the symmetry factor of 4. The calculations here are for $J = 0$ and thus vibration–rotation coupling is not a concern.

The number of open triplet channels at the singlet threshold depends strongly on the barrier height: $W_t = 114 (E_b = 4.4 \text{ kcal/mol})$ and $1037 (E_b = 0)$. Choosing a

4.0 kcal/mol barrier yields $W_t = 142$ at the singlet threshold. This value is close enough to the value $W'_i = 26$ extracted from the PST fit to the 0_{00} PHOFEX spectrum to provide a tenable though not fully satisfactory model. The RRKM calculation predicts that W_t should increase by about a factor of 2 over the $\approx 300 \text{ cm}^{-1}$ range of photolysis energies probed; the experimental data suggest a more rapid increase of W_t , Table II. This discrepancy may be due to theoretical⁴³ or experimental (Sec. V F) problems with deconvoluting W'_i from the PHOFEX curves.

This limit model is self-consistent if the internal conversion and intersystem crossing rates are much greater than the dissociation rates on the S_0 and T_1 surfaces. Because of its much smaller density of states the dissociation rate on the T_1 surface is much greater than that on the S_0 surface, at least in the energy region probed in the PHOFEX experiments. Using the experimental value W'_i , $k_T = 4 \times 10^{10} \text{ s}^{-1}$. For the approximations made here to be valid, k_{ic} and k_{isc} must be greater than $4 \times 10^{10} \text{ s}^{-1}$. This is also the condition for linewidth-state density products $\Gamma_{ic}\rho(\tilde{A})$ and $\Gamma_{isc}\rho(\tilde{A})$ to be greater than unity, which is required for complete mixing of the three electronic state manifolds. The corresponding limits on the rms matrix elements of the $|S_{1i}\rangle$ states are $|V_{ic}| > 0.0025 \text{ cm}^{-1}$ and $|V_{isc}| > 0.04 \text{ cm}^{-1}$.

B. The slow intersystem crossing limit ($k_{isc} \ll k_T$)

A different simple limiting case can also give reasonable agreement with all the experimental data. In this case the rate of formation of $^3\text{CH}_2$ is limited by the intersystem crossing rate rather than by the dissociation rate on the triplet surface. In this limit the coupling between the states

$$|\psi_m\rangle = \sum_i \alpha_{mi}|S_{1i}\rangle + \sum_j \beta_{mj}|S_{0j}\rangle \quad (34)$$

and the T_1 manifold is treated by perturbation theory. Each state $|\psi_m\rangle$ then decays irreversibly into the T_1 manifold at a rate $k_{isc}(m)$. Equations (30) and (31) can then be written

$$k_m = k_d \sum |\beta_{mj}|^2 + k_{isc}(m), \quad (35)$$

$$(\Phi_s/\Phi_t)_m = \frac{k_d \sum_j |\beta_{mj}|^2}{k_{isc}(m)}. \quad (36)$$

The summation in Eq. (36) must be less than but on the order of unity to be consistent with the experimental rise time and branching ratio measurements. This is also expected because the two singlet manifolds are fairly strongly coupled and $\rho(\tilde{X}) \gg \rho(\tilde{A})$. This model is self-consistent and in accord with experiment if $k_{isc} \approx 10^8 \text{ s}^{-1}$ (corresponding to an rms matrix element $|\langle \psi_m | H | T_{1k} \rangle|$ of 0.002 cm^{-1}). Intersystem crossing rates of this order of magnitude have been observed for other molecules.⁶⁷ Unfortunately, neither theoretical nor experimental intersystem crossing rates are available for ketene.

This model cannot explain the observed increase in W'_i with energy, Table II: the intersystem crossing rate is not expected to increase significantly over the energy range probed in the PHOFEX experiments since the $\tilde{a}$ and $\tilde{A}$ state geometries are almost identical.^{28,68} However, a significant

k_{isc} energy dependence may arise from the unknown vibronic structure of S_1 and T_1 . As discussed in Ref. 43, the increase in W' may be due to deviations of the singlet rate constant k_d from the PST calculation rather than an increase in the triplet formation rate.

It should be noted that the states $|\psi_m\rangle$ with the largest S_1 component and hence the largest oscillator strength are expected to have the smallest branching ratios in this model. Because of the similar geometries of S_1 and T_1 , the intersystem-crossing matrix elements should be much larger for $S_1 \rightarrow T_1$ than for $S_0 \rightarrow T_1$. The larger S_1 component in the wave function will increase the denominator in Eq. (36), and (slightly) reduce the summation in the numerator.

C. Summary

Two different physical models can explain the rapid switchover from the triplet to the singlet products as the total energy is increased. The first possibility is that the rate of formation of $^3\text{CH}_2$ is limited by dissociation over the barrier on the triplet surface, in RRKM fashion. In this model intersystem crossing (and internal conversion) must be rapid ($\geq 4 \times 10^{10} \text{ s}^{-1}$). Using *ab initio* transition state frequencies RRKM predicts a dissociation rate on the triplet surface 4 to 40 times larger than the $4 \times 10^{10} \text{ s}^{-1}$ inferred from the fits to the PHOFEX data. The second possibility is that the formation rate is limited by the intersystem crossing rate onto the triplet surface. This model can explain the data if the intersystem crossing rate is $\approx 10^8 \text{ s}^{-1}$. No *ab initio* or experimental estimation of this rate is available. The experimental data cannot distinguish between these two mechanisms. Both models fail to accurately predict the increase of the fitting parameter W' with energy. This may result from deviations of the singlet formation rate from PST away from threshold, as discussed in Ref. 43. From the PST fits to the PHOFEX data the rate of formation of $^3\text{CH}_2$ is estimated to be $1.6 \times 10^8 \text{ s}^{-1}$ at the singlet channel threshold, and the singlet-triplet branching ratio to rise from 0 to 2 in the first 125 cm^{-1} above threshold.

VII. CONCLUSION

The PHOFEX data yield a variety of important results. Many photofragmentation studies have been reported where product distributions are measured after photolysis at a fixed wavelength. Varying the photolysis wavelength, as in the PHOFEX studies reported here, can yield important complementary information, particularly in the threshold region. PHOFEX proves to be a very sensitive way to test competing theories of photofragmentation dynamics. PHOFEX also yields thermochemical information with spectroscopic accuracy.

The threshold measurements for different $^1\text{CH}_2$ rotational states yield a singlet channel dissociation threshold of $30\,116.2 \pm 0.4 \text{ cm}^{-1}$; combining this value with data in the literature yields $D(\text{CH}_2\text{--CO}) = 26\,960 \pm 6 \text{ cm}^{-1}$ and $\Delta H_{f0}^\circ(\text{CH}_2) = 93.6 \pm 0.4 \text{ kcal/mol}$. The differences in dissociation thresholds for the ortho and para nuclear spin states of methylene demonstrate that nuclear spin is conserved throughout the photodissociation, thus providing a

second system where nuclear spin conservation has been confirmed experimentally.

A model based on phase space theory (PST) is found to give excellent fits to the PHOFEX spectra. This agreement indicates that there are essentially no barriers to dissociation nor any significant dynamical biases among the accessible product channels. The calculation is not sensitive to the potential energy surface so long as it is sufficiently attractive to essentially remove the centrifugal barriers. The failure of the Prior model to reproduce the fine structure in the PHOFEX spectra demonstrates the importance of including the constraint of angular momentum conservation.

The statistical adiabatic channel model (SACM) and the related SACM-PST approximation fail to fit the data except in the limit where they become indistinguishable from PST (for $\alpha/\beta > 0.75$). Choosing the "standard" α/β ratio of 0.5 yields calculated curves which grossly deviate from the experimental PHOFEX spectra. It may be that the potential energy surface of ketene is unusually "isotropic"²⁰ or attractive at long range, and that more typical molecules will exhibit SACM behavior. However, the absence of any "adiabatic" barriers to dissociation suggests that the SACM theory may be physically incorrect, despite its successes at fitting other experimental data.^{4-7,11,20-25} More detailed work at low temperatures in the threshold region is called for on molecules, such as H_2O_2 ,^{7,11} which apparently behave as predicted by SACM. It may be that the phase-space contraction treated by SACM only occurs at higher product energies or higher angular momenta than those employed in the current study. PST, which works so well in the threshold region, has been shown to fail for several systems when product vibrations are accessed.¹⁴ Both SACM and the separate statistical ensembles (SSE) method⁸ correct for this deviation by treating product vibrations differently than rotations. Studies of ketene at higher photolysis energy will provide an interesting test case in the region where PST may prove inadequate. Very recent theoretical work by Wardlaw and Marcus,^{9,10} and by Troe²⁰ have aimed at unifying "tight" and "loose" transition state theories, by modifying older models. Additional results and discussion of the present work and its bearing on models for calculating dissociation rates and product energy distributions are presented in Ref. 43. The need for models to deal quantitatively with the strength of coupling of each coordinate to every other and the number of open channels, both as a function of distance along the reaction coordinate, presents an intimidating prospect.⁴³ It is hoped that the present work will help stimulate development of an accurate unified statistical theory of unimolecular dissociation.

From the PST fits to the PHOFEX spectra, the singlet-to-triplet branching ratio is seen to increase very rapidly with energy in the singlet threshold region, increasing from 0 to about 2 in the first 125 cm^{-1} above threshold. This rapid rise can be explained either by slow intersystem crossing or by slow dissociation over the barrier to form triplet methylene, relative to the rapid dissociation to form singlet methylene. The rate for unimolecular dissociation at the singlet threshold is estimated to be $\approx 2 \times 10^8 \text{ s}^{-1}$ from the PST fits and the calculated density of states. Direct measurement of

the CH_2 rise time at $30\,120\text{ cm}^{-1}$ yields $k > 7 \times 10^7\text{ s}^{-1}$. Experiments using picosecond lasers may be able to directly measure the absolute rate of unimolecular dissociation. This would provide additional tests of all the theories, and help clarify the relationship between dissociation rates and product state distributions.

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